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Studierichting: Informatica
Oefeningenreeks 4A nr. 43.9

$$I = \int \frac{x^2(x+2)}{x^4 - 2x^2 + 1} dx = \int \frac{x^2(x+2)}{(x-1)^2(x+1)^2} dx$$

$$\frac{x^2(x+2)}{(x-1)^2(x+1)^2} = \frac{A}{(x-1)^2} + \frac{B}{x-1} + \frac{C}{(x+1)^2} + \frac{D}{x+1}$$

$$A = \left.\frac{x^2(x+2)}{(x+1)^2}\right|_{x=1} = \frac{3}{4}$$

$$\left(\frac{x^2(x+2)}{(x-1)^2(x+1)^2} - \frac{3}{4(x-1)^2}\right)(x-1) = \frac{4x^2(x+2) - 3(x+1)^2}{4(x-1)(x+1)^2} = \frac{4x^3 + 8x^2 - 3(x^2 + 2x + 1)}{4(x-1)(x+1)^2}$$

$$= \frac{4x^3 + 5x^2 - 6x - 3}{(x-1)(x+1)^2} = \frac{(4x^2 + 9x + 3)(x-1)}{4(x-1)(x+1)^2} = \frac{4x^2 + 9x + 3}{4(x+1)^2}$$

$$\Rightarrow B = \left.\frac{4x^2 + 9x + 3}{4(x+1)^2}\right|_{x=1} = 1$$

$$C = \left.\frac{x^2(x+2)}{(x-1)^2}\right|_{x=-1} = \frac{1}{4}$$

$$\left(\frac{x^2(x+2)}{(x-1)^2(x+1)^2} - \frac{1}{4(x+1)^2}\right)(x+1) = \frac{4x^2(x+2) - (x-1)^2}{4(x-1)^2(x+1)} = \frac{4x^3 + 8x^2 - x^2 + 2x - 1}{4(x-1)^2(x+1)}$$

$$= \frac{(4x^2 + 3x - 1)(x+1)}{4(x-1)^2(x+1)} = \frac{4x^2 + 3x - 1}{4(x-1)^2}$$

$$\Rightarrow D = \left.\frac{4x^2 + 3x - 1}{4(x-1)^2}\right|_{x=-1} = 0$$

$$\Rightarrow I = \int \frac{3dx}{4(x-1)^2} + \int \frac{dx}{(x-1)} + \int \frac{dx}{4(x+1)^2}$$

$$= \frac{3}{4} \int \frac{dx}{(x-1)^2} + \int \frac{dx}{(x-1)} + \frac{1}{4} \int \frac{dx}{(x+1)^2}$$

$$= \frac{-3}{4(x-1)} + \ln|x-1| - \frac{1}{4(x+1)} + C$$

References